

Name: Key

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Please circle your section: 901 (Gossell) 902 (Gossell) 903 (Graves-McCleary)

No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. [8 points] Consider the function $f(x) = \frac{2}{3-x}$.

Find $f'(1)$ using the limit definition: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

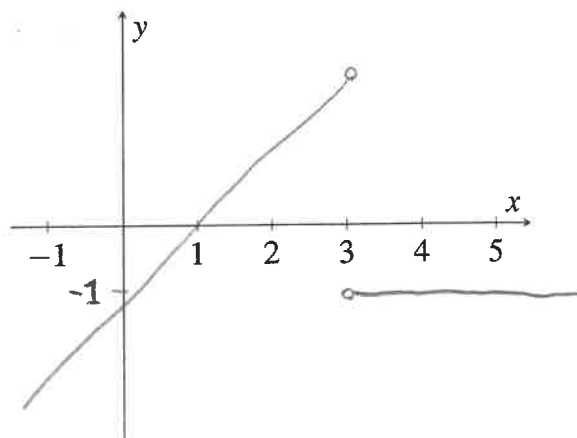
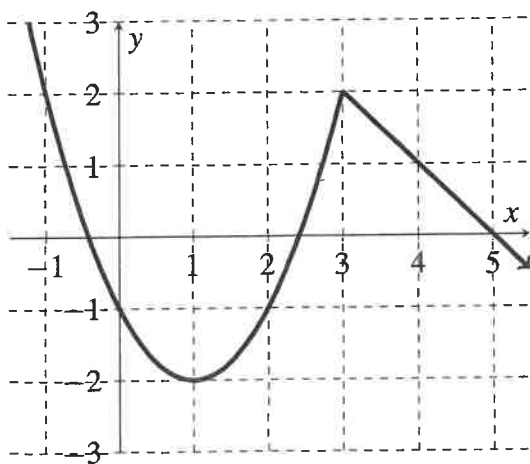
(Finding the derivative using any other derivative method will result in 0 points for this problem.)

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{\frac{2}{3-(1+h)} - 1}{h} = \lim_{h \rightarrow 0} \frac{2 - (3-(1+h))}{h(3-(1+h))}$$

$$= \lim_{h \rightarrow 0} \frac{2 - 3 + 1 + h}{h(2-h)} = \lim_{h \rightarrow 0} \frac{h}{h(2-h)} = \lim_{h \rightarrow 0} \frac{1}{2-h} = \frac{1}{2}$$

$$f'(1) = \boxed{\frac{1}{2}}$$

2. [5 points] Using the graph of $g(x)$ below, sketch a graph of the derivative $g'(x)$ on the axis to the right.



3. [12 points] Find the derivative of each function. You may use any method you like, and you do not need to simplify your answer.

a. $f(x) = 4x^7 - 5x^6 + \frac{x}{6} - 7$

$$f'(x) = 28x^6 - 30x^5 + \frac{1}{6}$$

b. $g(x) = \frac{30 + x - x^4}{x^2} = 30x^{-2} + x^{-1} - x^2$

$$g'(x) = -60x^{-3} - x^{-2} - 2x$$

c. $h(x) = \cos(x)(x^4 + x^3 + x + 1)$

$$h'(x) = -\sin(x)(x^4 + x^3 + x + 1) + \cos(x)(4x^3 + 3x^2 + 1)$$

d. $p(x) = \sqrt{x}(x^2 + \cos(x)) + 2^4$

$$p'(x) = \frac{1}{2}x^{-\frac{1}{2}}(x^2 + \cos(x)) + x^{\frac{1}{2}}(2x - \sin(x)) + 0$$